Pre Reading-Material

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These are some basic knowledge that may help you to understand the Sim Muscle practical. Please go through.

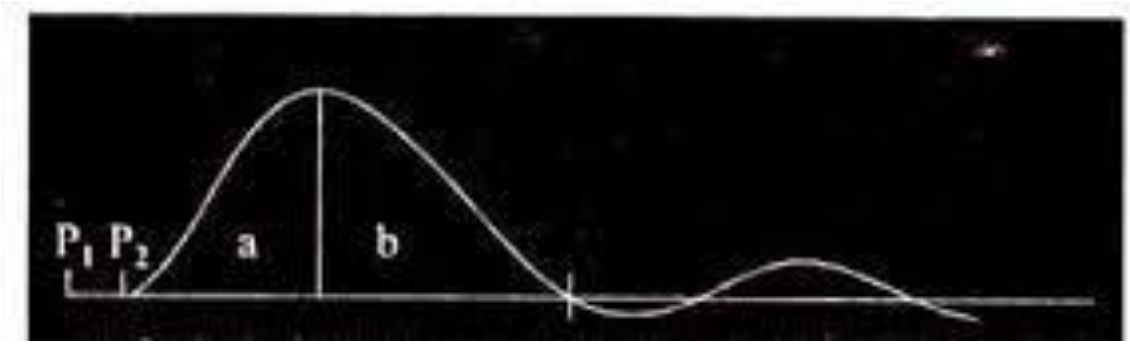


Figure: Simple muscle curve

P_1=Point of stimulation; P_2=beginning of contraction

P_1-P_2=latent period; a= contraction period; b=Relaxation period

The factors that can alter the excitability and the contractility of a simple muscle curve:

Strength and Duration of Stimuli:

A minimum strength and an adequate duration of the stimulus is necessary to excite a tissue. If you give a submaximal stimulus, the response will gradually be increased until it reaches to maximal stimulus. If maximal stimulus is provided one at a time, this will produce a single muscle curve with maximum force of contraction**.** It is true for either single muscle fiber or a whole muscle bulk. Generally, motor units are recruited based on their threshold (**Size principle**) when the stimulus intensity (voltage) is gradually enhanced (from low to high). The contraction period is more variable and depends on the properties of different motor units, whereas the relaxation period is primarily controlled by ATP-driven Ca^{2+} clearance into the sarcoplasmic reticulum. Relaxation rate also varies with muscle type (fast-twitch relaxes faster than slow-twitch), but this variation is generally less pronounced than that of contraction.

Effects of Two Successive Stimuli

If the second stimulus is applied after adequate intervals both first and second stimuli will cause contraction and two simple muscle curves will be recorded (Fig. See below). If the second stimulus is applied in the relaxation period of the first, two separate curves will be produced, and the second curve will be higher.

If the second stimulus is applied within the contraction period of the first one the second curve will be superimposed on the first one with a higher height of contraction (Fig. See

below). These are known as **summation** of effects (contraction). If the second stimulus is applied within the latent period but after refractory period of the first, then their effects are added together giving a simple muscle curve of larger height than either of them produced separately.

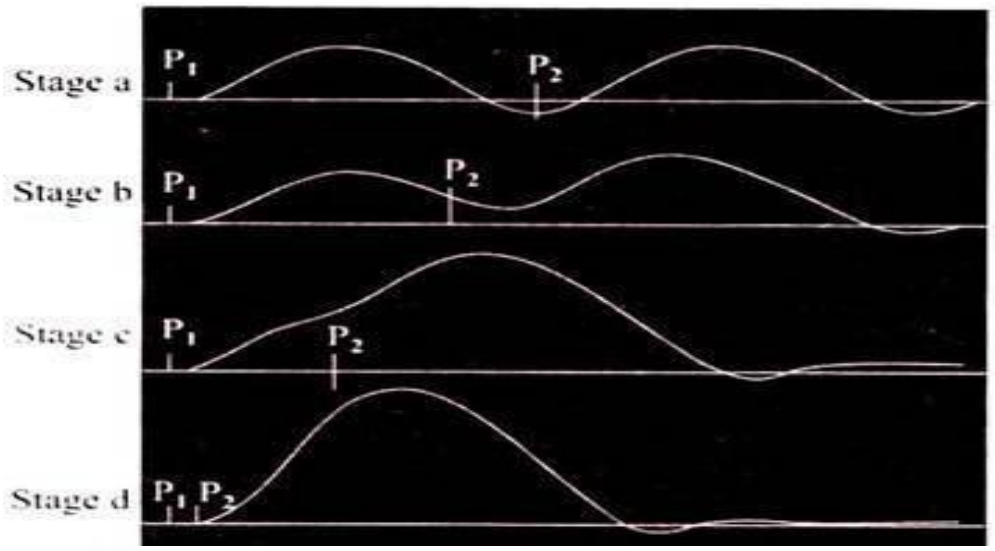


Fig: Effect of two successive stimuli at different points (a, b, c, d)

P_1= First point of stimulation; P_2=Second point of stimulation

Effects of Repetition of Stimuli (>2 or more):

When a series of such stimuli (5 to 6 stimuli) are applied but under the condition that each contraction can be completed before the next stimulus is applied, then a gradual increase in amplitude of contraction is obtained. This stair-like rise is called the **staircase (treppe) phenomenon** (See below figure A). Increased H^+ concentration and an increase of temperature within the muscle create a favorable condition for more work (beneficial effect).

When repeated stimuli is applied, the type of response will **vary according to frequency**. When the frequency is such that each successive stimulus falls within the period of relaxation of the previous curve the record will show a series of wavy oscillations. This is called **clonus or incomplete tetanus** (Figure below B-E)**.**

When the frequency is more, so that the stimuli fall within the latent period of the previous curve, the record traces a clear steady line, which rises at first abruptly and then gradually, till maximum contraction takes place. **This is called tetanus** where fusion is complete and the muscle, instead of vibrating, exerts a steady pull.

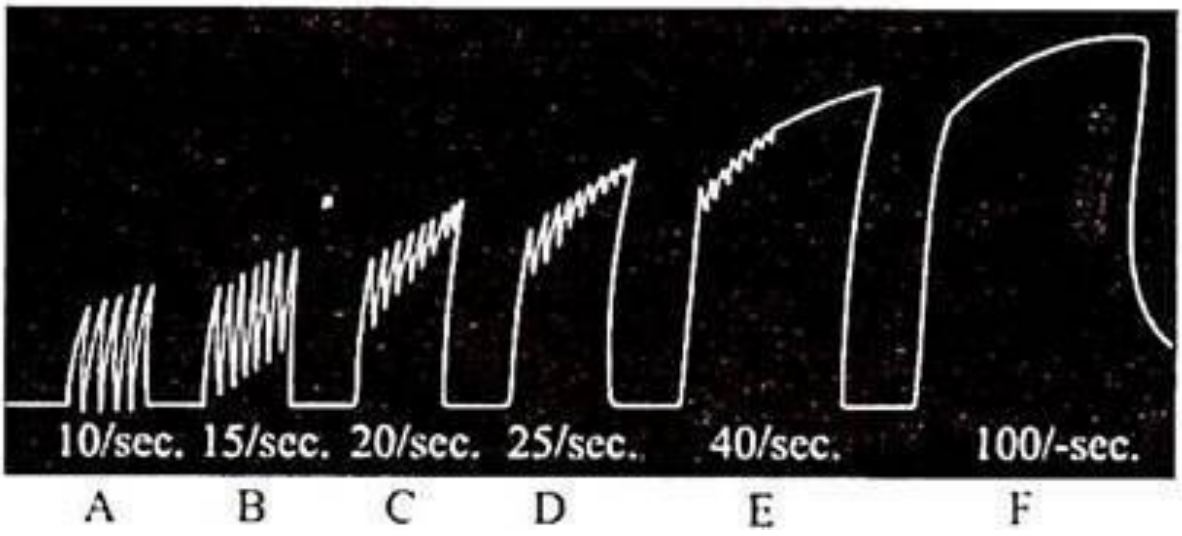


Fig: Genesis of tetanus

Effects of Load:

Load lengthens the latent period but reduces the periods of contraction and relaxation. It also reduces the degree of contraction, i.e., the height of the curve (Fig. see below). The effect of load on the work done by the muscle depends on the way in which the load is applied. If the weight is allowed to stretch the muscle prior to its contraction, the muscle is said to be **preloaded**. The preload enhances the ability to interdigitate more between actin & myosin molecules (ie: more cross bridge formation) as they are being pulled apart even before the contraction starts. However, this advantage could be seen only if the stretch remains within physiological limit. If it goes beyond the limit, muscle force will reduce instead.

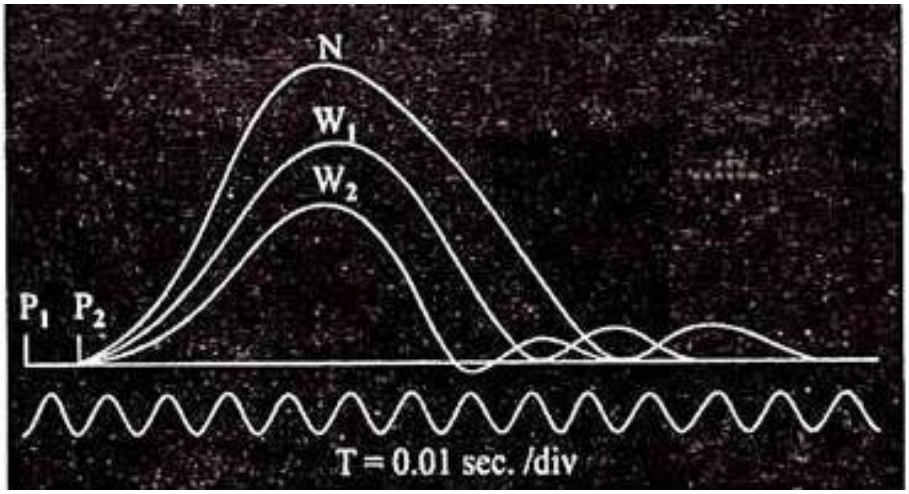


Fig: Effects of load

P_1=Point of stimulation; P_2=beginning of contraction

W1 and W2 are loads where W2> W1; N= Normal curve

Effects of fatigue:

When repeatedly stimulated, the muscle becomes gradually less excitable and ultimately ceases to respond. This phenomenon is called fatigue (Fig. See below). Muscular fatigue can be defined as the inability of the muscle to do further work. In the fatigue curve**, latent period is delayed, force of contraction diminished** along with the **relaxation period**.

The location of fatigue lies in the muscle when it is directly stimulated. But when it is stimulated through the motor nerve, the location could lie in **the neuromuscular junction**. In physiological exercise, the seat of fatigue is neither in the muscle nor in the neuromuscular junction but at the synapses in the central nervous system: called central fatigue.

When ATP becomes depleted after repeated stimulation, two things go wrong: myosin heads cannot detach from actin protein of the muscle, and calcium cannot be pumped back into the sarcoplasmic reticulum efficiently. Metabolic bi products also exert some influence. Both problems prolong relaxation. Additionally, any myosin heads still bound to actin are unavailable for subsequent contractions, which reduces both the rate and size of those contractions.

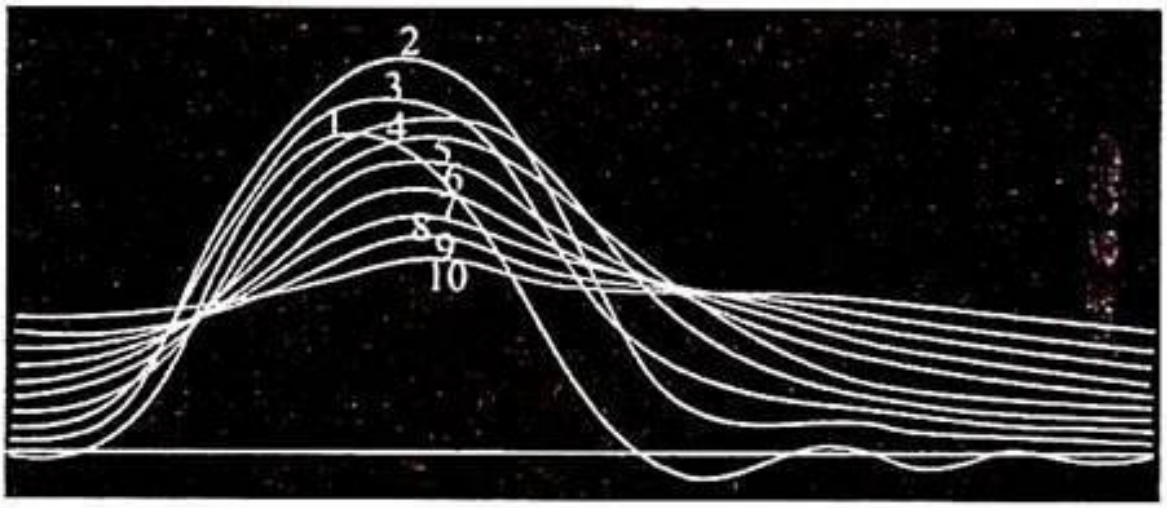


Fig: Effect of fatigue on simple muscle curve